

Chapter 3. Mechanics and Dynamics of Charting

Lecture notes | First semester

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What this chapter is about

These notes are the record of what Chapter 3 covered, written to be read on their own. If you were in the room, they are what to revise from. If you missed the session, they are the session. They follow the lecture in the same order and split into the same six parts, so you can move between the slides and these pages without hunting.

The chapter is about how a chart is built, so read it with the figures beside you. Every term is defined once, in the same words as the slides, and listed again at the back. The check yourself boxes are not assessed. Where the book leaves a question open, uses a word it has not defined, or says two different things, these notes say so rather than filling the gap from somewhere else, and Section 6.6 collects every one of those places.

After studying this chapter you should be able to:

1. Understand chart construction and how technical data is incorporated and displayed.
2. Describe the process by which OHLC data is created and its relationship to various charts.
3. Identify and differentiate between contango and backwardation, and their connection with negative and positive roll yields.
4. Understand the adverse effects of the bid-ask spread on trading performance.
5. Construct various charts using constant measures of time, range, volatility, trade volume and number of transactions.
6. Set up a volatility neutral chart for consistent viewing of price action.

Chapter scope follows Lim, M. (2016), *The Handbook of Technical Analysis*, chapter 3. Figures are reproduced from that text and remain the publisher's copyright.

Section 1

From Price to OHLC

What a chart is for and who reads one, the four families of data behind it, how a stream of trades becomes four prices, and why auto-scaling should be switched off.

1.1 What a chart is, and who reads one

Chapter 2 was a theory that kept only the closing price. This chapter builds the thing that theory is read from. The book describes traditional charting as a two dimensional matrix on which technical data is viewed.

A chart reveals repetitive price behaviour and market volatility, it shows up price distortions and illiquidity, and it is the surface on which trendlines, channels, envelopes and chart patterns are drawn.

Market data can be displayed as numbers or as a graph, and any numbers can be graphed. Quantitative analysts and statisticians work mainly with the numbers, using time series and stochastic analysis and back and forward testing, and for them a chart is optional. More traditional analysts prefer to work from the picture alone. **The book has a name for them.**

Chartist. Analysts who use charts to study technical data are called chartists. They study classical chart patterns, trendlines, window oscillators, overlay indicators and other price formations.

The picture a chartist works from is almost always the same shape: price up the side and time along the bottom. On a daily chart of a share, each step to the right is one more trading day and each step up is more pesos.

Price-time chart. The usual graphical form of technical data, where the vertical axis tracks the movement of price and the horizontal axis tracks the motion of time. The price axis may be scaled arithmetically (linear) or logarithmically (ratio).

One qualification about the time axis matters for the rest of the chapter. On some charts it is not plotted in equal units of time at all. **It acts instead as a counter for each new block of data, and on such charts time is regarded as implicit along the x axis.** Section 3 is full of charts that work this way.

The four families of technical data

- **Price data:** open, high, low and close.
- **Transaction related data:** volume and open interest.
- **Market breadth data:** advances, declines, total issues, up and down volume, new highs and new lows, and bullish percent data.
- **Sentiment data:** the put/call ratio, the short interest ratio, the specialist/public ratio, the cash/asset ratio, investor and advisor poll data, margin debt, and implied volatility (VIX).

The chapter lists all four families and then works only with the first two. **None of the breadth or sentiment items is defined in this chapter, and these notes do not define them either.** Where the book takes one up later, so will the course.

CHECK YOURSELF

On a chart whose time axis is only a counter, has time stopped passing? If not, what exactly has changed?

1.2 From a stream of trades to four prices

The majority of charts are simple price-time charts built from four prices per period, popularly called OHLC data. To make them, the continuous stream of trades is first cut into equal slices of time. Take one trading morning and cut it into five minute slices, 9:30 to 9:35, 9:35 to 9:40, and so on: each slice will become one bar.

Quantization of price. To create OHLC data, price activity is separated, or quantized, with respect to time, for each successive interval or period. An interval may be of any duration; the most popular are 1 minute, 5 minutes, 15 minutes, 1 hour, 4 hours, daily, weekly, monthly and yearly.

Quantize is the book's word, and filter is the word it uses beside it. Everything that happened inside the slice is then summarised as four numbers.

Suppose that in one five minute slice a share opens at 50.00 pesos, trades up to 50.80, down to 49.60, and ends at 50.40. Its OHLC is 50.00, 50.80, 49.60 and 50.40, and every other trade in those five minutes has been thrown away.

OHLC data. The price at the beginning and end of an interval is the opening (O) and closing (C) price; the highest and lowest prices within it are the high (H) and low (L). Price activity is summarized into these four pieces of information.

One five minute interval of jagged price filtered into four prices and drawn three ways: a 5 minute bar, a 5 minute candlestick and a 5 minute Gann bar, with the range marked.

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Figure 3.1 One interval of price filtered into four prices and drawn three ways.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.1 shows the filtering happen. On the left is the jagged price action inside one five minute interval; on the right are the same four prices drawn as a bar, as a Japanese candlestick and as a Gann bar. **The bar and the candlestick carry exactly the same information; only the drawing differs.** The chapter names Gann bars and draws one here, but never says how to read one, so nothing in this course is set on them.

Two more facts come with OHLC data. The range of a bar or candlestick is simply the absolute difference between its high and its low, $\text{range} = |H - L|$, so a bar with a high of 50.80 and a low of 49.60 has a range of 1.20 pesos. **And in most cases the closing price of one interval is also the opening price of the next, unless there is a gap in price.** Gaps are the second half of Section 2.

1.3 Long bars built from short ones

Once you have short bars, longer ones are built from them, not from the trades again. Take three five minute bars. The first opens at 50.00, the highest high of the three is 50.90, the lowest low is 49.40, and the third closes at 50.30. The fifteen minute bar they make is 50.00, 50.90, 49.40 and 50.30.

Higher timeframe bar. To create OHLC data over a longer period, take the opening price of the first period, the closing price of the last period, and the highest and lowest prices between them. Such composite bars are referred to as higher timeframe bars and candlesticks.

Figure 3.2 builds one 15 minute bar and one 15 minute candlestick out of three five minute intervals: the open of the first, the close of the last, and the extreme high and low wherever they fell. Figure 3.3 does the quantization for eight successive intervals and draws each bar and each candlestick underneath the price that produced it. Pick two neighbours and you can watch the close of one become the open of the next.

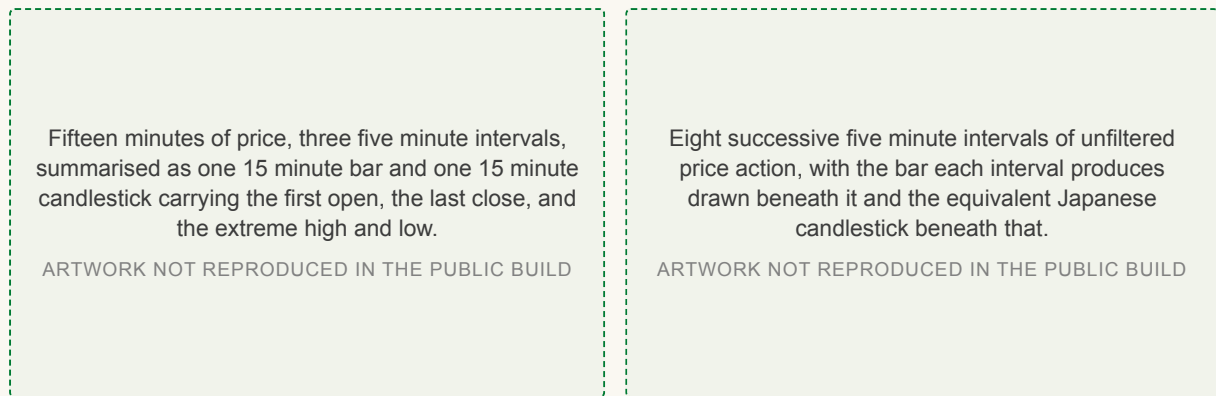


Figure 3.2. Three intervals, one bar.

Figure 3.3. Eight intervals, bar by bar.

Figures 3.2 to 3.3 A higher timeframe bar, and the quantization of a run of intervals.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

OHLC data is therefore simply a summary of price activity within an interval, and the longer the interval, the higher the timeframe of the bars it produces. Nothing new is added by going to a higher timeframe; the same trading is only filtered more coarsely.



Figure 3.4 OHLC data as the basis of most chart constructions.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.4 is a family tree with OHLC data at the top. Almost every chart in this chapter hangs off it: line, bar, Japanese candlestick, point and figure, three line break and Renko. The one exception has its own branch. **The equivolume chart also needs volume, so the book says it requires OHLCV data, where V is volume.** Section 3 comes back to that claim, because the book says something slightly different about it later.

The tree also names Kagi charts and Gann swing charts. They appear there and nowhere else in the chapter, which never explains either one, so neither is examinable from Chapter 3.

Figure 3.5 draws one sequence of OHLC data three ways, as a line chart, a bar chart and a candlestick chart. The line chart looks smoothest because it has thrown the most away, which Section 3 explains.

1.4 Switch auto-scaling off

Most charting software will, by default, stretch or squeeze the price axis so that whatever is on the screen fills it. Scroll a flat, quiet month into view and its bars suddenly look tall and wild; scroll a strong trend into the same window and those same bars shrink. **The bars did not change. Only the scale did.**

The same sequence of OHLC data drawn three ways: as a line chart, as a bar chart and as a candlestick chart.

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Figure 3.5 Three representations of the same price flow.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Auto-scaling. Auto-scaling attempts to fill the entire screen with price activity, so the heights of bars and candlesticks keep changing with whether price is ranging or trending. The vertical scaling is not preserved or constant, and volatility is misrepresented.

When prices are flat or ranging, auto-scaling stretches them across the screen and makes low volatility activity look more volatile. When prices are trending, it shortens every bar to fit the trend in and makes volatile activity look less volatile. Either way the vertical scaling is not preserved, so it becomes very difficult to learn what price behaviour at a given timeframe really looks like. **The book's instruction is plain: it is always best to turn auto-scaling off, because switching it off normalizes volatility on the charts and gives more accurate price visualization.** That is the answer to the chapter's second review question.

Four panels of the same prices: with auto-scaling on, an encircled quiet stretch looks volatile and then shrinks when a trend comes into view; with it off, the stretch keeps its true height.

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Figure 3.6 Auto-scaling misrepresenting volatility.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

In Figure 3.6 the top two panels have auto-scaling on. The encircled quiet stretch is portrayed as more volatile than it is while the screen is flat, and then suddenly shrinks once a later trend has to be fitted in. The bottom two panels have it off, and the encircled stretch keeps its true height.

One honest note. The chapter's learning objectives ask you to set up a volatility neutral chart. **That phrase never appears in the chapter's text; switching off auto-scaling, which the book says normalizes volatility, is the nearest it comes.**

CHECK YOURSELF

Three five minute bars are 50.00, 50.60, 49.70, 50.20, then 50.20, 50.90, 50.10, 50.70, then 50.70, 50.75, 49.40, 50.30. Write down the fifteen minute bar before you look back.

Section 2

What the Four Prices Mean, and Four Kinds of Gap

Why the high and the low carry more weight than the open and the close, the three conditions that make the open and close matter more, and the four ways the book measures a gap.

2.1 Two prices set by the clock, two by traders

OHLC data is the result of filtering price over an interval, but the four prices do not carry the same significance. **The open and the close are merely a function of time stamping: they mark where an arbitrary interval happened to start and stop, and once the interval ends, the price at that instant becomes the close of one bar and usually the open of the next.**

The high and the low are different in kind. They are created by actual market forces of supply and demand. **The book's own figure labels the high a function of supply and the low a function of demand: sellers capped the high, buyers held the low.** A share trades up to 52 pesos and sellers meet it there; it trades down to 49 and buyers meet it there. Both prices were set by people risking money, not by the clock.

Price rejection. The highs and lows represent areas of price rejection, caused by the responsive actions of market participants, who react to high and low prices by risking capital. This makes them more significant than the open and close, and the longer the interval, the more significant the high and low formed.

One period of price with the open and the close each labelled a function of an arbitrary time stamp, the high labelled a function of supply and the low a function of demand, both as market forces.

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Figure 3.7 What sets each of the four prices.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.7 labels one period's four prices by what sets them: time stamp, supply, time stamp, demand. The final sentence of the definition is the practical one. **A weekly high means more than a five minute high, because the longer the interval, the more significant the high and low it forms.**

2.2 When the open and the close matter more

Having demoted the open and the close, the book gives three conditions under which they gain importance and attention from market participants, and become more reliable and actionable. They are the answer to the chapter's third review question.

The three conditions

1. The durations between active trading sessions are longer.
2. The opening and closing prices belong to a higher timeframe bar or candlestick.
3. There is a larger price gap between the previous close and the new open.

Three trading sessions, each with its own open, high, low and close, separated by stretches of no trading, annotated that longer breaks raise the significance of the open and close.

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Figure 3.8 Longer breaks between sessions raise the significance of the open and close.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.8 is the first condition in a picture: three trading sessions separated by stretches of no trading. **The longer the market has been shut, the more its opening price means.** That is why the daily open and close draw more interest than the opens and closes between the morning and afternoon sessions of one day: the break overnight is longer than the break at lunch. A daily open is also a higher timeframe price than a five minute one, so it meets the second condition at the same time.

The other side of the same argument is a market with no break at all. **Spot foreign exchange trades continuously through the week, from Sunday evening to the Friday close, and in continuously traded markets the open and close matter less than the high and the low.**

CHECK YOURSELF

Why should a daily closing price be taken more seriously than the close of a five minute bar in the middle of the morning? Give two of the three conditions.

2.3 Four ways to measure a gap

A share closes at 102 pesos and opens the next morning at 106. Nobody traded at 103, 104 or 105 in between: price jumped over them.

Gap. A gap is represented by a range of prices where no trading activity takes place. There are essentially four ways to define one, Types 1 to 4.

The four types are four ways of measuring the same jump, and they are easiest to learn on one pair of bars. Say the last bar had a high of 104 and closed at 102, and the new bar opens at 106, then trades between a low of 105.50 and a high of 108.

Types 1 and 2 are measured to the new bar's open, so both exist the moment the market opens: close to open is 4 pesos, and high to open is 2.

Type 1 and Type 2 gaps. Type 1 is measured from the close of the previous bar to the open of the next or current bar. Type 2 is measured from the high or low of the previous bar to the open of the next or current bar. Both are created instantaneously at the open, with no trading within that range.

Types 3 and 4 are measured to the new bar's high or low, so you only know them once the new bar has closed: previous high to new low is 1.50 pesos, and previous close to new low is 3.50. For a gap downward, read low for high and high for low; that is why every definition says high or low.

Type 3 and Type 4 gaps. Type 3 is measured from the highs or lows of the previous bar to the highs or lows of the current bar, and is the gap usually called a window in candlestick and bar charts. Type 4 is measured from the close of the previous bar to the high or low of the next. Both are created only after the current bar has closed.

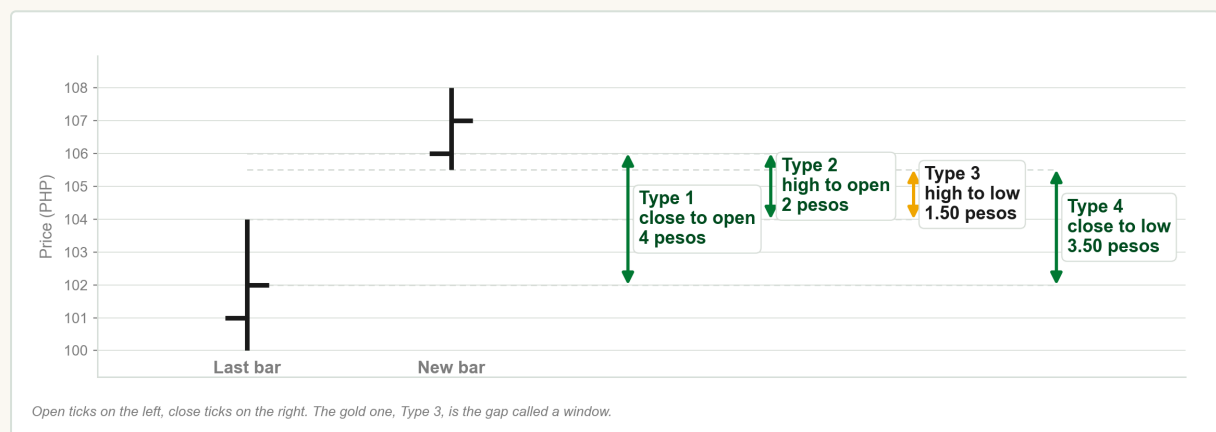


Chart A The four gaps measured on one pair of bars, in pesos.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart A draws exactly that pair of bars with the four brackets measured between them. Follow each bracket back along its dashed guides to see which two prices it joins. The gold one is Type 3, the gap usually called a window in Japanese candlestick and bar charts: the two bars do not overlap at all.

Four schematic pairs of bars, Types 1 to 4, each gap marked by an arrow; Types 1 and 2 grouped as created when the next bar opens, Types 3 and 4 as needing the next bar to close.

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Figure 3.9 Four definitions of a gap.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.9 is the book's schematic of the same four, grouped the useful way: Types 1 and 2 created instantaneously when the next bar opens, Types 3 and 4 requiring the next bar to close before the gap can be identified.

2.4 What a gap does afterwards

A gap normally represents an area of support or resistance, depending on whether price is above or below it, and this is especially so for a Type 3 gap. Larger gaps are more significant. **Prices are generally expected to return and fill the gap at a later date, although there are many instances where this does not occur.** Read both halves of that sentence: the book does not say that gaps always fill.

The chapter uses support and resistance here without defining them, exactly as Chapter 2 did. Chapter 5 is where the book defines them, and these notes do not borrow a definition in the meantime.

The book also notes that the average true range (ATR) is somewhat related to a Type 4 gap. The true range of a bar is the greater of its own range and the distance from the last close to its high or low. On the running example the new bar's range is 108 minus 105.50, which is 2.50, but the distance from the last close of 102 to the new high of 108 is 6, so its true range is 6 pesos. **The chapter gives the true range and not the average; how ATR is averaged is taught in Chapter 8.** ATR comes back in Section 3, as one way to size a constant volatility bar.

Finally, the book looks ahead. Chapter 5 analyses gaps in relation to market phase, and the four it covers are common gaps, breakaway gaps, runaway gaps (also called continuation or midway gaps) and exhaustion gaps. Those are names for later. **Do not confuse the two lists: Types 1 to 4 are ways of measuring a gap, and the Chapter 5 list is kinds of gap.**

CHECK YOURSELF

The last bar closed at 60 with a low of 58. The new bar opens at 55 and trades between 53 and 56. Measure all four gap types for this gap downward.

Section 3

Five Constant Measures

The five things that can decide when a bar is finished, the four constant time charts, the charts that ignore the clock, and the one chart with no constant at all.

3.1 What decides when a bar is finished

Every chart has to decide when one bar ends and the next begins. Every bar has a range, a duration, a volume, a number of transactions and a volatility, and each kind of chart holds one of those constant and lets the rest vary. The book calls them the five types of constant measure chart, and this list is the answer to its fourth review question.

The five constant measures

1. **Constant time:** the bar closes when a specified time interval is met. Candlestick, bar, equivolume and line charts.
2. **Constant range:** the bar closes when a specified excursion in price is met. Point and figure and Renko charts.
3. **Constant volume:** the bar closes when a specified volume is met.
4. **Constant transaction**, or tick: the bar closes when a specified number of transactions is met.
5. **Constant volatility:** the bar closes when a specified amount of standard deviation or ATR is met.

3.2 Constant time charts

On a daily chart, a quiet day and a wild day each get exactly one bar of the same width. That is the defining property of the commonest kind of chart.

Constant time chart. Constant time charts quantize price activity into units of time, so each bar is complete once a specified amount of time has elapsed; they are also called interval charts. The bar duration is constant, and the range, volume, transactions per bar and volatility are variable.

Constant time charts are the most popular form of chart construction, and the book gives a reason that matters later. It sorts overlay indicators into three kinds and treats them fully in Chapter 8: geometric ones drawn from points on the chart, such as trendlines, channels and chart patterns; numerical ones computed from the prices, such as moving averages; and horizontal ones, such as prior support and resistance levels. Constant time charts work with both numerically and geometrically based overlays. Charts that are not constant time should use numerical and horizontal overlays only, because their time axis is not linear, and a line drawn between two points depends on how far apart the points are.

The first constant time chart draws each interval as one vertical line. A day that opened at 50, ran up to 52, fell to 49 and closed at 51 is a line from 49 to 52, with a small tick on the left at 50 for the open and a tick on the right at 51 for the close, which is how the book's own figures draw it.

Bar chart. A bar chart is created by the quantization of price into specified time intervals. Bar charts clearly depict levels of supply and demand in the market, but the open and close markers are hard to read from a distance when the chart holds many bars.

The second throws nearly all of that away. Five days closing at 50, 51, 50.50, 52 and 53 pesos become five dots joined by a line, and the highs and lows of those days appear nowhere.

Line chart. Line charts are created by connecting all of the closing prices. They give little information about levels of supply and demand, since they ignore the high and low prices, which are the true indicators of market force. They are used as a trend identifier or a simple summary of price activity.

Chart B shows the loss. The pale bars are each day's high and low; the joined dots are the line chart. The line never reaches the extremes, and those extremes are precisely the prices Section 2 called the true indicators of market force.

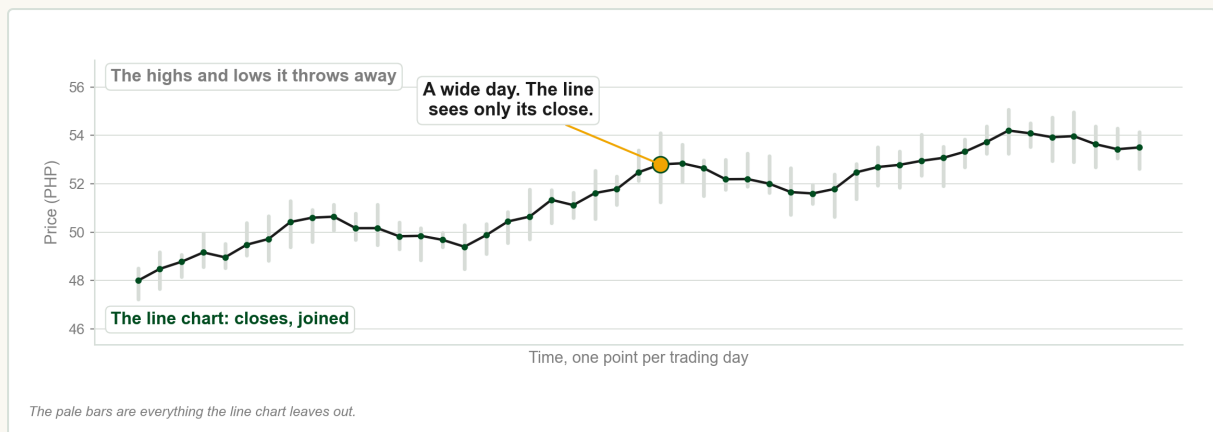


Chart B A line chart drawn over the highs and lows it leaves out.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

So a line chart filters out more price information than a bar chart, which is why it says little about supply and demand and is still useful as a trend identifier. It is, in effect, the chart Chapter 2's Dow Theory reads.

The third constant time chart is a bar with the stretch between the open and the close drawn as a box. Open 50 and close 51 is a hollow, or white, box from 50 up to 51. Open 51 and close 50 is the same box filled, or black.

Japanese candlestick. Japanese candlesticks are created in essentially the same manner as price bars. The only difference is that the space between the opening and closing price is boxed up and referred to as the real body, which may be filled or hollow, black or white, depending on where the open lies relative to the close.

A white bullish candlestick and a black bearish one labelled part by part: real body, open, close, and the upper and lower shadows, also called tails or wicks, running to the high and low.

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Figure 3.10 Bullish and bearish Japanese candlesticks.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.10 labels both kinds part by part: the real body between open and close, and the upper and lower shadows, also called tails or wicks, running out to the high and the low. Candlestick formations may be classified as bullish or bearish, reversal or continuation, and simple, double or multiple; Chapter 14 takes them further. **One limit applies to bars and candles alike: neither can tell you whether the high or the low came first, because the intrabar detail has been filtered out.**

The fourth constant time chart puts volume inside the bar. Two days have the same high and low; one traded 3 million shares and the other 1 million. On an equivolume chart the first is drawn as the wider bar.

The book says only that larger volume gives a wider bar, not in what proportion, so these notes say no more.

Equivolume chart. Equivolume charts are based on constant time bars, displaying volume over the interval as bar width rather than height. They are constructed using high and low prices, and open and close prices are disregarded. Larger volume results in a wider bar.

Two things about equivolume are traps. **The first is that every bar still covers one interval of time, so it is a constant time chart, and yet its time axis is not plotted in a linear fashion, because the bars are different widths.** Trendlines, channels and chart patterns are therefore distorted on it, while numerical and horizontal overlays are not.

The second is that the book gives two accounts of its data. Beside Figure 3.4 it says equivolume bars require OHLCV data. Here it says they are constructed using the high and the low, with the open and the close disregarded. **Learn both statements, and know that the bar itself draws on the high, the low and the volume.** No assessment in this course will turn on the difference.

The same daily prices of Apple Inc. as a candlestick chart with a linear time axis and as an equivolume chart with a non linear one, the trendline on each ending at a different price.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.11 A candlestick chart and an equivolume chart of the same prices.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.11 draws the same daily prices of one stock both ways. Look along the bottom of each: even spacing on the candlestick chart, uneven spacing on the equivolume chart. Then look at where the two trendlines end, at different prices, although they were drawn on the same data.

CHECK YOURSELF

Which of the four constant time charts would you use to see supply and demand, and which to see only the trend? Why?

3.3 Constant range charts

Set a size of 1 peso. A share that drifts 40 centavos all week draws nothing at all; a share that moves 3 pesos in a morning draws three. **Time has dropped out of the chart completely.**

Constant range chart. On constant range charts each bar, box or brick is complete once a specified excursion in price is met. The bar range and the bar volatility are constant; the duration, volume and transactions per bar are variable.

The most popular constant range chart draws rising prices as a column of X's and falling prices as a column of O's.

With a box size of 1 peso and a reversal size of 3 boxes, price must move at least 1 peso to add a box in the direction of the column, and at least $3 \times 1 = 3$ pesos against the column before a reversal can be plotted in a new column.

Point and figure chart. The most popular form of constant range charting. A continuation box is the minimum excursion for a new box in the trend's direction; the reversal size is the number of boxes needed to plot a reversal. The box size is arbitrarily chosen, and closing prices are normally used, though many use the high and low.

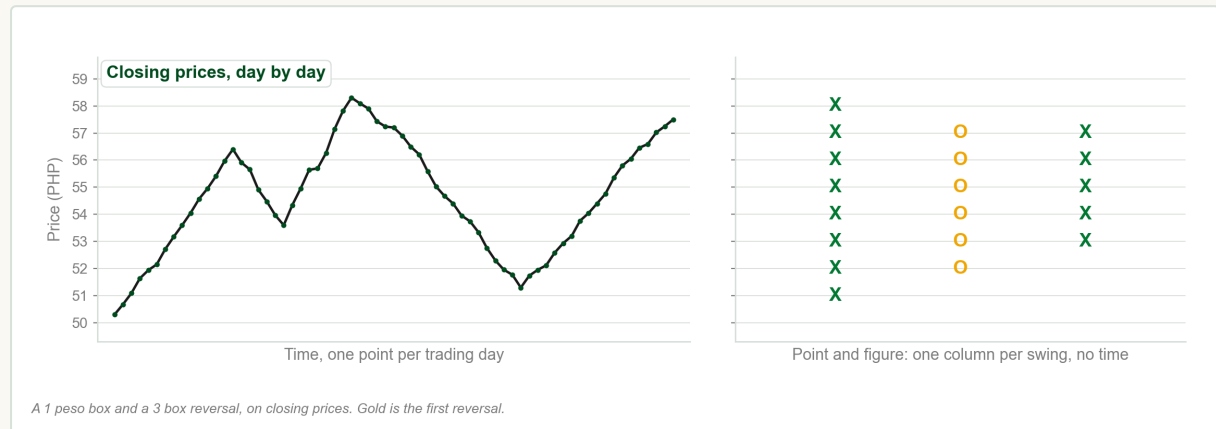


Chart C Point and figure columns built from a closing price path, 1 peso box, 3 box reversal.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart C builds the columns from a price path, using closing prices as the book says is normal. Follow the left panel with a finger and compare it with the right. The first dip is too shallow to reverse the X column, because it is less than three boxes; the second one is exactly deep enough, and the gold O column starts. Notice that the right panel has no time axis at all: one column per swing, however many days the swing took.

That missing time axis has a consequence. Because the time axis is non linear, geometrically based overlays would give inconsistent readings, so point and figure charting uses its own kind of trendline. **It needs only one point, and rises and falls along 45 and minus 45 degree angles.** The book calls these bullish support and bearish resistance lines, used in place of conventional trendlines.

A point and figure chart of Google Inc. in columns of X's and O's, with a plus 45 degree line under the rise and a minus 45 degree line over the decline, and the note that the time axis is non linear.

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Figure 3.12 A point and figure chart with its 45 degree lines.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.12 is a point and figure chart of one stock with a plus 45 degree line under the rise and a minus 45 degree line over the decline. Read the date labels along the bottom and see how unevenly they are spaced.

The second constant range chart uses bricks. With a brick size of 1 peso, a rise from 50 to 52.40 adds two white bricks, and the 40 centavos over 52 do nothing. For a black brick, price must move at least two bricks, 2 pesos, the other way.

Renko chart. A form of constant range charting whose bars are bricks, bullish white and bearish black, each plotted in a new column once the minimum price excursion is met. All reversals require price to move at least two bricks the other way, so Renko charts are essentially two brick reversal charts. Brick size is arbitrary.

A Renko chart of the iPath Dow Jones-AIG Coffee Total Return Sub-Index in white and black bricks, with the note that its time axis is non linear.

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Figure 3.13 A Renko chart, with its non linear time axis.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.13 is a Renko chart of a commodity index in white and black bricks. **Compare it with point and figure: every brick gets a column of its own, and a reversal needs two bricks, which is why the book calls Renko charts essentially two brick reversal charts.**

3.4 Volume, transactions and volatility

The next two charts count trading instead of price. Set 100,000 shares a bar, and a busy morning might draw ten bars while a slow afternoon draws one.

Constant volume chart. In constant volume charts a new bar is plotted once the minimum volume traded is met. The bar volume is constant; the range, duration, transactions per bar and volatility are variable. Since a new bar is not time dependent, the time axis is plotted in a non linear fashion.

The book gives no figure of a constant volume chart, so Chart D is ours. It draws one session's price and volume on an ordinary time axis and drops a gold line wherever another fixed block of shares has traded. **Every gold line marks the same volume; only the time between them changes.** The lines crowd together in the busy hours and spread out in the quiet ones, and that spacing is exactly what a non linear time axis means.

A tick chart counts trades rather than shares. Set 50 trades a bar, and fifty trades of 100 shares and fifty trades of 10,000 shares each make exactly one bar.

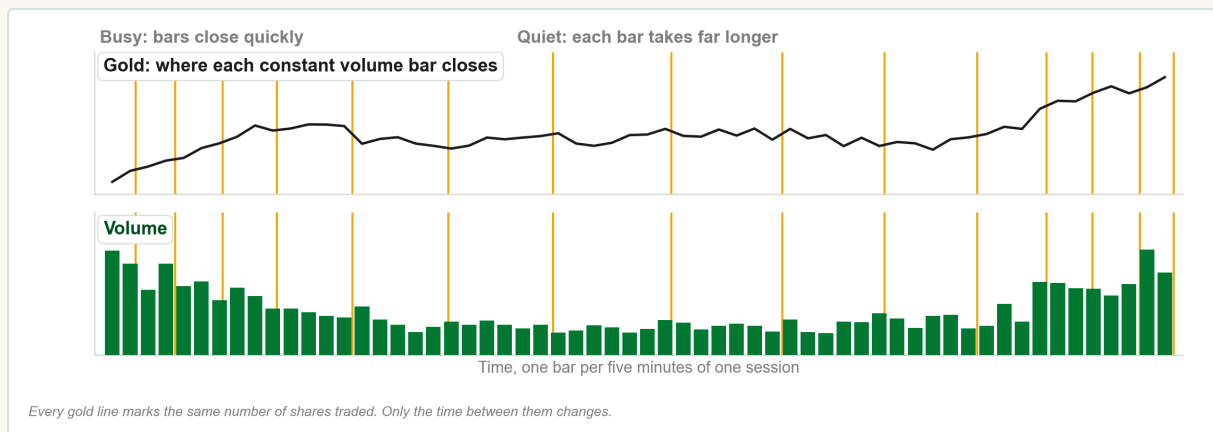


Chart D Where each constant volume bar would close in one session.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Constant transaction chart. In constant transaction or tick charts a new bar is plotted once the minimum number of transactions is met. Each transaction is represented by one tick, and the volume per transaction is unspecified. The transactions per bar are constant; everything else is variable, and the time axis is non linear.

The last constant measure lets volatility set the size of each bar. Set each bar at 2 x ATR: if the ATR is 0.50 pesos a bar needs a 1 peso move, and if the ATR widens to 1.50 it needs 3 pesos. It resembles a constant range chart, except that the minimum price excursion is determined by volatility rather than fixed.

Constant volatility chart. A new bar is plotted once a minimum price excursion determined by volatility is met, for example 2 x ATR. The bar volatility is constant; range, duration, volume and transactions are variable. It is objective, as volatility sets the size, and subjective in the choice of the ATR multiple.

Two point and figure charts of one market: box size set to 1 x ATR on the left, less volatile and more directional; standard box sizing on the right, more volatile and less directional.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.14 Constant volatility against constant range charting.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.14 draws the same market twice as point and figure: once with the box size set to 1 x ATR, which the book annotates as less volatile and more directional, and once with standard box sizing, annotated as more volatile and less directional. The book's point is that the volatility based version identifies trends more effectively. **It is objective, because the market's volatility sets the size, and subjective in the choice of how many multiples of ATR to use.**

3.5 The chart with no constant at all

Three line break chart. Three line break charts possess no measures of constancy. Once price closes above the previous high or below a previous low, a new line is created. After three successive lines, a reversal may only be plotted if price meets or exceeds the low, or in an upside reversal the high, of the last three.

After three white lines in a row, price has to fall to the low of those last three lines before a black line can be drawn. The book calls the three line break chart unique, because it is the only construct in the chapter with nothing held constant: range, duration, volume, transactions and volatility all vary. Bullish lines are white and bearish lines black, as with Renko bricks.

The same period of the Currency Shares Euro Trust drawn as a Renko chart and as a three line break chart side by side.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.15 Three line break and Renko charts of the same market.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.15 puts a Renko chart and a three line break chart of the same period side by side. The big turns appear on both; the spacing of the same months does not match.

Which one each chart holds constant

- **Constant time:** the duration.
- **Constant range:** the range, and the volatility.
- **Constant volume:** the volume.
- **Constant transaction:** the number of transactions per bar.
- **Constant volatility:** the volatility.
- **Three line break:** nothing at all.

Everything not named on its line is variable. **One oddity is worth noticing rather than resolving: the book lists volatility as constant for both constant range and constant volatility charts.** What separates them is the range, which is fixed on one and set by volatility on the other.

CHECK YOURSELF

On which of these is the time axis non linear: a bar chart, an equivolume chart, a Renko chart, a tick chart? Explain the one that surprises people.

Chart Scaling

Three ways to scale a price axis and what each preserves, which overlays scaling changes and which it leaves alone, and why the same prices can look bullish on one scale and bearish on another.

4.1 Three ways to scale a price axis

The book names three scales: linear or arithmetic, ratio or logarithmic, and square root. The first two are the common ones and Chapter 2 already used them under their arithmetic and logarithmic names. This section gives the arithmetic behind them.

On a linear scale, a rise from 10 to 20 pesos and a rise from 90 to 100 pesos are drawn the same height, because both are 10 pesos. But the first is a 100 percent rise and the second is only 11.1 percent, 10 divided by 90.

Linear scaling. In linear or arithmetic scaling, equal distances on the chart represent equal price changes: the change $P(N) - P(N-1)$ is constant for every equal distance moved on the chart. The percentage change is not preserved for equal units of price change.

A ratio scale makes the opposite trade. A rise from 10 to 20 pesos is 100 percent, and on a ratio scale it is drawn the same height as a rise from 90 to 180 pesos, which is also 100 percent even though it is a 90 peso move.

Ratio scaling. In ratio or logarithmic scaling, equal distances on the chart represent equal percentage changes: $(P(N) - P(N-1)) / P(N-1)$ is constant for every equal distance moved, which is also what is called stock returns. The price change is not preserved per equal unit of percentage change.

One sentence in the book contradicts this definition. Two sentences after defining ratio scaling as equal distances for equal percentage changes, it says that equal distances on the chart do not equate to equal percentage changes. The definition is stated twice and drawn twice, and the stray sentence is neither, so learn the definition. No assessment in this course rests on the stray sentence.

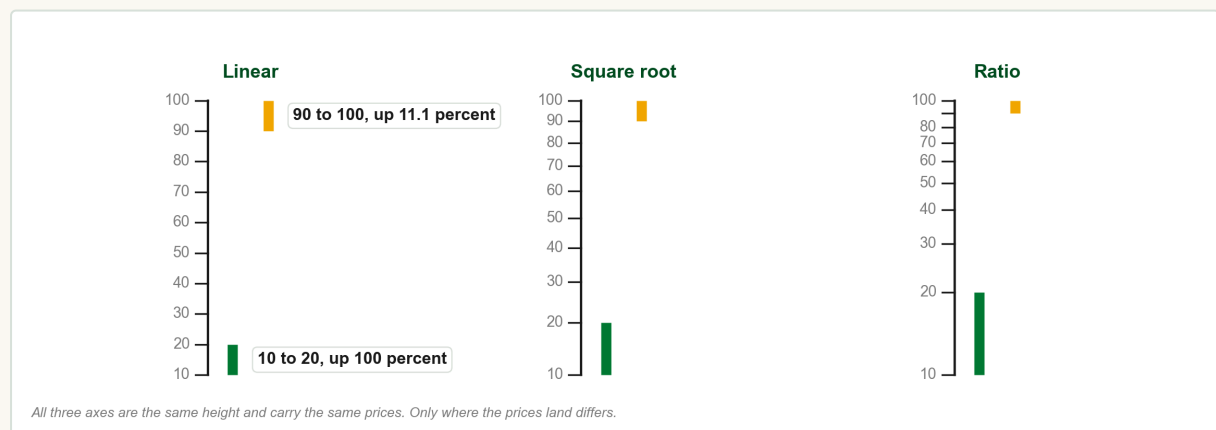


Chart E The same prices on a linear, a square root and a ratio axis.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart E places the same prices, 10 to 100, on all three axes, each drawn to the same height, and marks the book's two moves on each. On the linear axis the two bars are the same height.

On the ratio axis the rise from 10 to 20 is far taller than the rise from 90 to 100. The square root axis falls in between, which is the whole of what the book says about it.

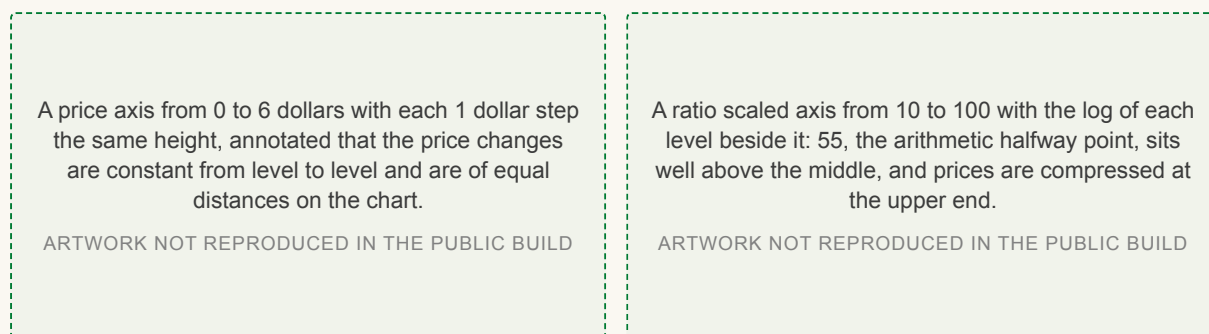


Figure 3.16. Linear.

Figure 3.17. Ratio, and the halfway point.

Figures 3.16 to 3.17 Equal price steps, and where 55 lands on a ratio scale.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.16 shows a linear axis in equal one dollar steps. Figure 3.17 shows a ratio axis from 10 to 100 with the logarithm of each level beside it: 55, the arithmetic halfway point, has a log of 1.74, above the midpoint of 1.50, so it sits high on the axis. **That is compression at the upper end, in numbers.**

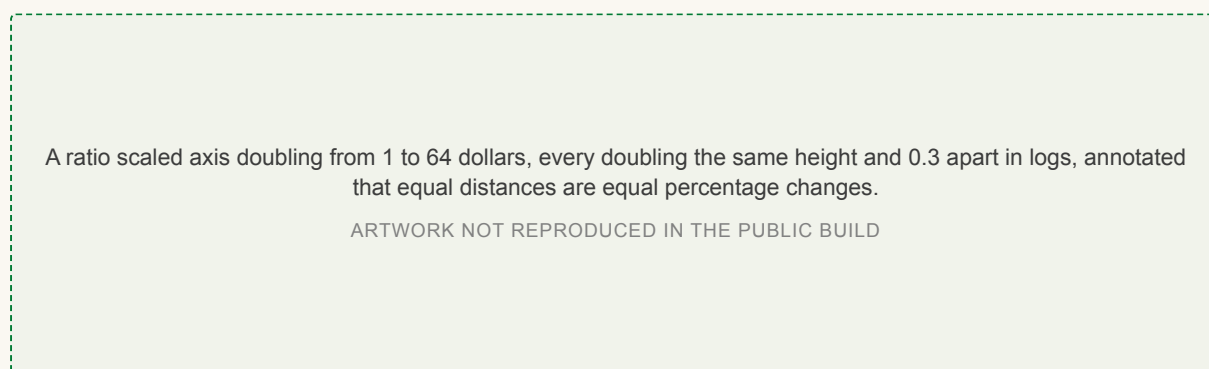


Figure 3.18 Equal percentage steps on a ratio scale.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.18 makes the same point the other way round: an axis doubling from 1 to 64, with every doubling drawn the same height and 0.3 apart in logarithms.

4.2 Which scale to use

The ratio scale compresses price action at higher prices and expands it at lower prices, while linear scaling spreads price evenly across the whole range. So linear scaling gives better definition and visualization at the upper end of prices and poorer clarity at the lower end, and ratio scaling gives better definition at the lower end and poorer clarity at the upper end.

The book's general guideline follows from that. Use linear charts when the price range under observation is relatively small. For stock price ranges exceeding 100 dollars it may be more appropriate to use ratio charting, and ratio charts are best for very long term market action, mainly in equities. **For futures, foreign exchange, and any instrument traded on very low margin, that is very high leverage, linear charts are more suitable, because ratio charts would compress price too much at higher prices.** The book does not say whether its 100 dollar threshold means the price itself or the distance from low to high, and these notes do not guess.

Square root scale. The square root scale lies somewhere between the linear and ratio scales with respect to its scale increments, and is not commonly used on most platforms. It indicates an uptrend line penetration later than ratio scaling but earlier than linear scaling.

Equal price steps on a square root axis get shorter as price rises, but not as much shorter as on a ratio axis. **The book calls its in between signal rather disconcerting: most traders react to uptrend line violations on either a linear or a ratio chart, so a trader using square root scaling will enter and exit either too early or too late relative to the levels everyone else is watching.**

CHECK YOURSELF

You are charting a highly leveraged currency pair over three months. Which scale does the book recommend, and why not the other one?

4.3 What scaling changes, and what it does not

This is the answer to the chapter's fifth review question. **Scaling affects every overlay that relies on geometry in its construction: trendlines, channels and chart patterns.** Change the scale and the angle of every trendline changes with it, so it indicates different levels of potential support and resistance. Scaling does not affect overlays constructed numerically, such as moving averages and price envelopes, and it does not affect overlays that mark horizontal levels: prior support and resistance, Fibonacci extensions, retracements, projections and expansions, Gann one eighth and one third retracements, and Gann Square of Nine projections. A horizontal line is the same price on any scale. The chapter only names these overlays; each is taught later in the book.

When trendlines are penetrated

- On **ratio** charts, uptrend lines are penetrated sooner and downtrend lines later.
- On **linear** charts, uptrend lines are penetrated later and downtrend lines sooner.

The SPDR Dow Jones Industrial Average on an arithmetic and on a logarithmic scale side by side, with the uptrend line penetrated later on the arithmetic chart and sooner on the logarithmic one.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.19. Uptrend lines.

The U.S. Natural Gas Fund on an arithmetic and on a logarithmic scale side by side, with the downtrend line penetrated sooner on the arithmetic chart and later on the logarithmic one.

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Figure 3.20. Downtrend lines.

Figures 3.19 to 3.20 The same trendlines broken at different times on the two scales.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.19 draws the same uptrend on an arithmetic and a logarithmic chart, and the line breaks first on the logarithmic one. Figure 3.20 does the same for a downtrend, and there the arithmetic chart breaks first. Chapter 2 gave the uptrend half of this rule; this is the whole of it.

Switching scales can change the whole mood of a chart. **The book gives four cases, and they reduce to one sentence: ratio makes both directions look bearish, and linear makes both look bullish.**

The four appearances

- Rising prices on a **ratio** chart may decelerate in a concave curve, like a flattening out or rounding top: a potentially bearish appearance.
- Declining prices on a **ratio** chart may accelerate in a convex curve, like a downward parabolic move: a potentially bearish appearance.
- Rising prices on a **linear** chart may accelerate, or rise evenly, in a convex curve, like an upward parabolic move: a potentially bullish appearance.
- Declining prices on a **linear** chart may decelerate in a concave curve, like a flattening out or rounding bottom: a potentially bullish appearance.

The same decline in A123 Systems Inc. on a semilog scale, where it takes a convex, parabolic and bearish appearance, and on a linear scale, where it takes a concave and more bullish appearance.

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Figure 3.21. A decline.

The same advance in Apple Inc. on a ratio chart, where it appears to flatten out with a bearish appearance, and on a linear chart, where it does not flatten and appears bullish.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.22. An advance.

Figures 3.21 to 3.22 The same prices looking bearish on one scale and bullish on the other.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.21 is a decline drawn on a semilog scale, another name the book uses for ratio, where it takes a convex, parabolic and bearish appearance, and on a linear scale, where it looks concave and more bullish. Figure 3.22 is an advance that appears to flatten out on a ratio chart and does not flatten on a linear one. **The prices did not change their meaning; the scale supplied the mood.**

Energy Services of America Corp. on semilog and linear charts: channels drawn from the same inflection points land in different places, while the moving average reads the same on both.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.23 Channels on linear and ratio scaling.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.23 closes the argument on channels. The same inflection points produce channels in different places on a semilog and a linear chart, while the moving average reads the same on both: geometric overlays are affected by scaling, numerical ones are not.

CHECK YOURSELF

A share has risen steadily for five years. On which scale is it more likely to look as though it is rounding over, and would that be a fact about the share?

Section 5

The Bid-Ask Spread

Why most charts are drawn on the bid and what that does to a buyer, the three problems of the expensive, early and late longs, and how the spread bends the reward to risk ratio.

5.1 The price on the chart and the price you pay

The book's worked examples in this part are in dollars. These notes use pesos; the arithmetic is identical. Suppose the bid, the price you can sell at, is 10.00 pesos, and the ask, the price you must pay to buy, is 10.20. The spread is 20 centavos, and every trade that buys and then sells pays it.

Bid-ask spread. The bid-ask spread is the price difference between the bid and the ask price. A buy must accept the ask price and a sell is made at the bid. Since a trade consists of a buy and a sell, the bid-ask spread affects all trades.

Charts are typically drawn on bid prices, although many platforms can also draw price on the ask or on the mid price. The book's rule is that on a bid-based chart, trade performance is adversely affected when initiating long entries and executing long exits, and that it does not specifically affect short entries or short exits.

That rule needs one reading note. The book defines shorting, in this part, as selling to open or close a position. **Read long the same way, as any buy, whether it opens a position or closes one.** Read like that, the rule says buying is hurt and selling is not, which is what the book's own examples show. Read long as the position rather than the order, and the rule would contradict the example that follows it.

A price rising to a 10.00 dollar resistance level on a bid-based chart, the ask 20 cents above at 10.20, and the short entered at the bid at exactly the level intended.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.24 Shorting resistance at the bid.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.24 is the easy case. Resistance sits at 10.00 on the bid chart, price tests it, and you sell at market. **You are filled at 10.00, slippage aside, because selling happens at the bid, which is the price the chart shows.** A sell limit order at 10.00 fills there or better, and the same is true when the sell is an exit, a stop loss or a profit target.

5.2 Three ways a buyer gets the wrong fill

Now buy instead. The breakout level is 10.00 and you plan to buy at 10.05. Price touches 10.05 on the bid chart, you buy at market, and you are filled at the ask: 10.25. **Right time, wrong price.**

Expensive longs. The problem of the expensive longs: a long initiated with a market order at precisely the right time on a bid-priced chart is nevertheless bought at a higher price than intended, because a buy must accept the ask price.

A price breaking up through a 10.00 dollar resistance level, with the long entered as price reaches 10.05 on the bid chart and filled at the ask of 10.25, twenty cents above.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.25 Going long at the breakout.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.25 shows the two lines that make the problem: the chart said 10.05 and the account says 10.25, twenty cents above, which is the spread you paid.

Use a pending order to get the price exact and the problem moves rather than going away. A buy stop at 10.05 fills at 10.05, the ask, but at that moment the bid is 9.85, still below the 10.00 breakout level on the chart. **Right price, wrong time.**

Early longs. The problem of the early longs: a buy stop entry order is filled at the intended price, the ask, while the bid is still below the breakout level, so the trader has bought into the position before the breakout on the bid-priced chart.

Support is the mirror case. Buy at market as price tests a 10.00 support level and you are filled at the ask, 10.20: the expensive longs again. Use a pending order to buy at 10.00 instead and you are filled at 10.00, but only once the bid has already fallen to 9.80, below the support level. **At a breakout the pending order is early; at support it is late.**

Late longs. The problem of the late longs: a pending order to buy at a support level is filled at the intended price, the ask, only when the bid is already below the support level on the bid-priced chart.

Figure 3.26 shows the late fill: the ask at 10.00 and the bid at 9.80, the long filled at the ask by a pending order once the bid is already below the level. **The figure labels its level a resistance level, while its own title and the text call it support.** It is support, and these notes read it that way.

CHECK YOURSELF

With a 20 centavo spread, name the problem in each case: a market buy at a breakout, a buy stop at a breakout, and a pending buy at support.

Price falling back to a 10.00 dollar level, ask 10.00 and bid 9.80, the long filled at the ask by a pending order once the bid is below the level. The figure labels the level resistance.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.26 Going long at support.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

5.3 The spread and the reward to risk ratio

The spread also bends the odds of every trade. It increases the probability of exiting at the stop loss, which the book calls loss promoting, and it reduces the probability of exiting at the profit target, which it calls profit restricting. How it shows depends on the kind of order.

With a market order, the spread is deducted from the profit and added to the loss, and it is reflected in the trading account. A trader buys when the bid is 1.00; the spread is 0.20, so the trader is long at 1.20. With a profit target at 1.50, the most the trade can make is 0.30, not 0.50. With a stop at 0.50, the most it can lose is 0.70, not 0.50. **The spread adjusted reward to risk ratio is 0.30 divided by 0.70, which is 0.43, not 1.**

A long bought when the bid was 1.00 dollar and filled at the ask of 1.20, stop 0.50 and target 1.50: expected R/r of $0.50/0.50 = 1$ against a spread adjusted $0.30/0.70 = 0.43$.

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Figure 3.27. A market order.

A long bought at the ask of 1.00 dollar by pending order while the bid is 0.80, stop 0.50 and target 1.50: R/r stays 1, but price must rise 0.70 to win and fall only 0.30 to lose.

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Figure 3.28. A pending order.

Figures 3.27 to 3.28 The spread against market and pending long entries.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.27 sets that out, the expected ratio of $0.50/0.50 = 1$ beside the spread adjusted $0.30/0.70 = 0.43$. Figure 3.28 is the pending order version, and there the cost is hidden. The spread is not deducted from the profit or added to the loss, and it does not show in the account: the ratio stays $0.50/0.50 = 1$. But the order was filled at the ask of 1.00 while the bid was 0.80, so price must rise 0.70 from the bid to reach the target and fall only 0.30 to reach the stop. **Same ratio on paper, worse odds in the market.**

The remedy is to make the spread small relative to the trade: increase the size of the profit target and the stop loss relative to the spread, and the obvious way to do that is to trade a higher timeframe or a larger wave cycle. A 20 centavo spread is small against a 5 peso target and most of a 30 centavo one. **That is why the spread hurts the short term trader most, and the scalper above all.** Scalpers need to buy at the bid and sell at the ask to reduce their costs, and a standard level one (L1) trading platform will not let a trader do either.

This part answers the chapter's seventh review question.

CHECK YOURSELF

Bid 2.00, spread 0.10, bought at market. Target 2.40, stop 1.80. Work out the expected and the spread adjusted reward to risk ratios.

Section 6

Futures Contracts

Why futures exist and why they expire, which contract to trade, rolling over at a premium or a discount, contango and backwardation, and three ways to chart a string of contracts.

6.1 Contracts that expire

Futures contracts were originally created to let commodity producers hedge against falling prices. By shorting an equal amount in the futures market, a producer need not worry about prices falling at harvest time, having locked in production costs and any potential profit whatever prices do afterwards. **That is why the book calls futures essentially a bearish mechanism.**

The chapter describes what futures are for and how they behave, and it never defines a futures contract itself. These notes do not supply a definition from elsewhere. Everything in this section follows from one fact the book does state: unlike shares, every futures contract eventually expires, so a position has to be moved from one contract to the next.

Rollover. Unlike the equity markets, all futures contracts eventually expire, requiring a trader who intends to hold a position to roll over into the next available contract. Expiration months are usually quarterly, in March, June, September and December, often extending beyond a year.

A contract is normally fairly illiquid through most of its life, until the last three to six months before expiry. **Volume and open interest are greatest about two to three months before expiry, and as expiry approaches, volume subsides as traders roll into the next contract.** That decline can mislead an analyst into thinking a trend is weak, when a quick look at the next contract would show volume rising over the same period. To gauge the true volume and open interest across rollover points, the book says to use a continuous chart.

Daily charts of the July and September 2011 silver futures side by side: volume contracting on the expiring July contract while it expands on the September contract, which has become the nearby one.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.29. Two contracts.

The continuous daily chart of silver with its volume, giving one balanced view of volume action across multiple contracts and marking a volume blow off.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.30. One continuous chart.

Figures 3.29 to 3.30 Volume on two silver contracts, and on the continuous chart.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.29 shows it happening: volume contracting on the expiring July silver contract while it expands on the September contract, which has become the nearby one. Figure 3.30 is the continuous chart of the same months, one balanced view of volume across several contracts.

Falling volume on an expiring contract is traders leaving, not the trend weakening. The practical advice follows: trade the contract with the largest volume and open interest, which may not be the nearby one, and roll slightly before expiry to avoid the volatility of traders exiting at the last moment.

6.2 Naming the contracts

It is March and the March contract has just expired. June is now the contract that expires soonest.

Front month contract. The contract with the closest expiry is the nearby, nearest or front month contract. It has the expiration date closest to the current date, the narrowest bid-ask spread, the narrowest spread with respect to the spot price, and it is the most liquid contract.

The next contract is the second one out, with the expiry closest to the nearby contract; with June as the front month, September is next. **Contracts further out than that, December and the following March, are back month contracts.** Traders may keep rolling into the next nearby contract, may roll only into the next month contract two contracts further out, or may hold one contract closer and another further out at once to create a calendar spread or for other hedging purposes.

Quarterly contracts over twelve months, March to March, showing for each starting point which contract is the nearby or front month, which is the next month and which is the back month.

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Figure 3.31 Identifying the front, next and back month contracts.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.31 lays out quarterly contracts over twelve months and shows, for each starting point, which contract is the front month, which is the next month and which is the back month. Drop down one row and every label moves one contract along.

Delivery months are identified by letters: F January, G February, H March, J April, K May, M June, N July, Q August, U September, V October, X November and Z December. **The quarterly four are H, M, U and Z.**

CHECK YOURSELF

List the four characteristics of the front month contract, and say which way each of the two spreads runs.

6.3 Rolling at a premium or a discount

The new contract may not be trading at the same price as the one expiring. Sell March at 11 pesos and buy June at 12, and you pay 1 peso to stay in the trade.

Negative roll yield. When the new contract trades at a higher price, it is trading at a premium to the previous contract, and rolling over at a premium causes the trader to experience negative roll yields. It is possible to lose money being long in a rising or sideways market if the rollover premiums are large enough.

Reverse the prices, selling March at 12 and buying June at 11, and you collect 1 peso for staying in the trade.

Positive roll yield. When the new contract trades at a lower price, it trades at a discount to the previous contract, and rolling over at a discount causes the trader to experience positive roll yields. It is possible to make a profit being long in a declining or sideways market if the rollover discounts are large enough.

A long bought in March at 10 dollars and sold at 11 when March expires, then rolled into the June contract at 12, paying a 1 dollar premium: a negative roll yield.

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Figure 3.32. At a premium.

A long bought in March at 10 dollars and sold at 12 when March expires, then rolled into the June contract at 11, collecting a 1 dollar discount: a positive roll yield.

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Figure 3.33. At a discount.

Figures 3.32 to 3.33 Losses from negative roll yields, and profits from positive ones.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figures 3.32 and 3.33 are the two cases drawn side by side, each a long bought in March at 10 dollars and rolled into June. In Figure 3.32 the roll pays a 1 dollar premium; in Figure 3.33 it collects a 1 dollar discount. **Successive rolls accumulate, which is how the two surprising sentences in the definitions come about: a long position can lose money in a rising market, and make money in a falling one, if the premiums or discounts are large enough.**

6.4 Contango, backwardation and convergence

Contracts further out may trade at successively higher or lower prices. Suppose the spot price is expected to be 100 at expiry, and the contracts further out trade at 101, 102, 103 and 104.

Normal contango. When further out contracts are trading at successively higher prices compared to the expected spot price at expiry, the market is in normal contango.

Now suppose they trade at 99, 98, 97 and 96 against the same expected spot price of 100.

Normal backwardation. If further out contracts are trading at successively lower prices compared to the expected spot price at expiry, the market is in normal backwardation.

Traders often drop the word normal and compare the further out contracts with the nearby contract or the current spot price instead. The book calls that an informal use of the terms, but common practice among futures traders. **The difference matters: simple contango can be read off a chart, while normal contango needs the expected spot price at expiry.** The book does not say how to estimate that expected price; it says only that some knowledge of it is required.

A daily chart of 2012 gold futures, with the further out contracts stacked beneath it trading at successively higher prices than the nearby December contract.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.34 Simple contango in 2012 gold futures.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.34 shows gold futures with each further out contract trading higher than the nearby December one. That is simple contango, and the book notes that it may or may not be normal contango. The next fact ties the two ideas together.

Convergence. Futures prices always converge to the spot price at expiry. With the spot price constant, a contract in contango will eventually decline, experiencing negative yields, and a contract in backwardation will eventually rise, experiencing positive yields.

Contracts converging on a spot price assumed to stay at 100 dollars: backwardated prices of 99 to 96 rising to it with a positive yield, contango prices of 101 to 104 falling with a negative yield.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.35 Futures and spot prices converge at expiration.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 3.35 draws it with the spot assumed to stay at 100: the backwardated contracts at 99 to 96 rise to meet it, with a positive yield, and the contango contracts at 101 to 104 fall to meet it, with a negative yield. With some mispricing, the book adds, a trader may go long a far out backwardated contract and short a nearby one and lock in the gain from convergence whatever the direction of price, although the spread and trading costs still decide whether it pays.

The book's real example of negative roll yield is natural gas in 2009. Natural gas rallied strongly through the year, but the U.S. Natural Gas Fund, which tracks gas through futures and so has to roll, rolled at relatively large premiums and lost money. **Contango can take a whole rally away from a holder who has to roll.** Very short term traders who trade individual contracts and never roll are neither hurt by negative roll yields nor helped by positive ones.

Figure 3.36 puts the rallying continuous chart of natural gas beside the fund over the same months, and Figure 3.37 is the fund's relative strength against spot gas, falling steadily and labelled as the effect of negative roll yield.

The continuous daily chart of natural gas, in contango and rallying, beside the U.S. Natural Gas Fund over the same months.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.36. Gas and the fund.

A relative strength chart of the U.S. Natural Gas Fund against spot natural gas, falling steadily and labelled as the effect of negative roll yield.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.37. The fund relative to gas.

Figures 3.36 to 3.37 Natural gas in contango against the fund that tracks it.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The book does not say what happened to the fund afterwards, and neither do these notes. Together this subsection is the answer to the chapter's sixth review question.

CHECK YOURSELF

Why does a contract in contango tend to fall toward expiry even if the spot price does not move at all?

6.5 Charting a string of contracts

Between the nearby and the next contract there may be a premium or discount gap. For a trader it is only a payment made or received at rollover. For the chartist it is a problem, because a chart needs a continuous record of price and the contracts do not join up. The book gives three popular approaches.

The simplest is to join the nearest contracts end to end and leave the jumps in. March expires at 110 and June is trading at 120, so the chart simply jumps 10 at the roll.

Unadjusted nearest futures chart. Connecting the nearest contracts without accounting for the spread between them. It preserves the actual sequence of nearby prices and is an accurate record of historical prices, but technical analysis cannot be applied to it meaningfully, and back and forward testing are impossible because of the gaps.

A price series made of successive contracts joined end to end without adjustment, with a price gap distortion at every rollover.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.38 An unadjusted nearest futures chart.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 3.38 shows the jumps, a price gap distortion at every rollover. Each jump is a roll, not a market move, which is why a trendline drawn across them means nothing. **Accurate history, useless analysis.**

Back-adjusted continuous chart. Back adjusting raises or lowers the previous contract price at expiry, or N days before, to match the new contract, removing the spread and creating a continuous flow of prices. It always displays the current market price and allows technical analysis and back and forward testing.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Price (PHP)

Adjusted history below zero

Grey: each contract joined as it traded

Black: the same prices back adjusted

zero

Time, one point per trading day, five contracts in turn

Every new contract here starts at a discount to the last, the case in which the book warns of this.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Figure 3.39 is the effect on a real market: the continuous, spread adjusted chart of gold above the December 2013 gold contract, with the OHLC readings for the same day boxed on each. They differ, because the continuous chart's history has been rewritten by adjustment and the contract's has not.

The continuous, spread adjusted daily chart of gold above the December 2013 gold contract over the same period, with the OHLC readings for 7 November 2011 boxed on each and differing between the two.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 3.39 A continuous chart against the contract itself.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

The shifting history causes one more problem. Relative or comparative strength charts will not display accurate ratios if past prices keep changing every time the data is back adjusted. For that job the book prefers the third approach.

Perpetual contract. Perpetual contracts are not back shifted with each new front month contract. They display an estimated price rather than the current market price, interpolated between expiration dates by a weighting factor. That makes them useless for trading, but useful as a basis for constructing relative strength charts.

Between two expiries, a perpetual series blends the nearby and further out contract prices by a weighting, so the value on the chart is not a price anyone traded at. The book does not give the weighting, and these notes do not supply one.

Which chart for which job

- **Trading, and back or forward testing:** the back-adjusted continuous chart, which always displays the current price.
- **An accurate record of what actually traded:** the unadjusted nearest futures chart.
- **Relative strength charts:** perpetual contracts, because back adjusting keeps shifting the ratio.

None of the three does every job. The list above, with Figure 3.40, answers the chapter's eighth review question.

6.6 What this chapter names and does not teach

A chapter about how charts are built uses a great many words it does not stop to explain, and it is worth collecting them in one place. **None of the following is examinable from Chapter 3, and none of it should be filled in from outside the book.**

Gann bars, Kagi charts and Gann swing charts are named and drawn but never explained. The breadth and sentiment data items are listed and never defined. The average true range is used without its averaging, which Chapter 8 teaches. Support, resistance, trendlines and channels are used throughout and defined in Chapter 5. Normal contango needs an expected spot price the book does not say how to estimate. The ratio chart guideline gives a 100 dollar threshold without saying what it is measured on. The learning objectives ask for a volatility neutral chart, a phrase the text never uses. And a futures contract is never defined at all.

The book also contradicts itself twice and mislabels one figure, and all three are named where they occur: one stray sentence against the definition of ratio scaling, two accounts of the data an equivolume bar uses, and a support level labelled resistance in Figure 3.26. **In each case the right answer is to say what the book says and to mark the edge of it, rather than to borrow a tidier answer from somewhere the examiner is not reading.**

Chapter 3 in five sentences

Summary

1. A chart filters each interval into four prices: open, high, low and close.
2. Supply and demand set the high and low; the clock sets the open and close.
3. A bar can close on time, range, volume, transactions or volatility.
4. Scaling moves every trendline, and the spread moves every entry.
5. Futures expire, each roll costs or pays, and no futures chart does every job.

Review questions

These are the chapter's own review questions. Every one is answered somewhere in the six sections above, and they are the shape the quiz takes.

1. How is OHLC data derived, and what increases its significance?
2. Why should traders normalize their charts for volatility?
3. Describe the five measures of constancy in charting.
4. How does chart scaling affect the use of overlay indicators?
5. How does the bid-ask spread affect the reward to risk ratio?
6. Link contango to negative roll yields; describe back-adjusted charts.

Key terms

Every term this chapter defines, against the subsection that defines it.

Chartist	1.1	Constant volatility chart	3.4
Price-time chart	1.1	Three line break chart	3.5
Quantization of price	1.2	Linear scaling	4.1
OHLC data	1.2	Ratio scaling	4.1
Higher timeframe bar	1.3	Square root scale	4.2
Auto-scaling	1.4	Bid-ask spread	5.1
Price rejection	2.1	Expensive longs	5.2
Gap	2.3	Early longs	5.2
Type 1 and Type 2 gaps	2.3	Late longs	5.2
Type 3 and Type 4 gaps	2.3	Rollover	6.1
Constant time chart	3.2	Front month contract	6.2
Bar chart	3.2	Negative roll yield	6.3
Line chart	3.2	Positive roll yield	6.3
Japanese candlestick	3.2	Normal contango	6.4
Equivolume chart	3.2	Normal backwardation	6.4
Constant range chart	3.3	Convergence	6.4
Point and figure chart	3.3	Unadjusted nearest futures chart	6.5
Renko chart	3.3	Back-adjusted continuous chart	6.5
Constant volume chart	3.4	Perpetual contract	6.5
Constant transaction chart	3.4		

Sources

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